

Question			Marking Details	Marks Available					
				AO1	AO2	AO3	TOTAL	Maths	Prac
4	(a)	i	Number of bubbles (in a minute for 5 minutes)		1		1		1
		ii	The glucose is used up (in respiration) / {decrease in live population of yeast/ some yeast die}/ Build up of toxic waste NOT reference to oxygen		1		1		1
		iii	Bubbles vary in size(1) measuring <u>volume</u> of gas (1) OR Bubbles too fast to count (1) slow motion recording / measuring <u>volume</u> of gas (1)		1	1	2		2
		iv	Decrease/ become acidic (1) CO ₂ {dissolved in solution/released/ produced}/ carbonic acid produced (1)			2	2		1
		v	Sucrose is {a disaccharide/ made of glucose and fructose} (1) Has to be {hydrolysed/ broken down} (before it can be respired)/ 3 minutes was not enough time {to produce sucrase / for sucrase to act}/OWTTE (1)		2		2		
		vi	Any two × (1) from: Can be used in all organisms (1) For {most/ all} reactions (1)	2			2		
	(b)	i	Any three × (1) from: Glucose is used in glycolysis takes place in the cytoplasm (1) Glucose cannot enter mitochondria/ pyruvate can enter the mitochondria(1) Pyruvate used in link reaction and takes place in the mitochondria (1) no enzymes for glycolysis in mitochondria/ ORA (1)		3		3		
		ii	ADP required to generate ATP (1) Accept equation Oxygen is {final electron acceptor/terminal electron acceptor/ converted to water} (1)		2		2		

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	(c)	i	Released as heat (1) Keeps baby warm/ maintains body temp/ counteracts high surface area to volume ratio (1)	1	1		2		
		ii	Higher number of capillaries (1) to supply the tissue with {oxygen / nutrients / named nutrient}/ remove carbon dioxide from the tissues (1) and distribute the produced heat throughout the body/ increased respiration rate/ produces more heat to maintain body temperature (1)		1	2	3		
			QUESTION 4 TOTAL	3	12	5	20	0	6